

Infrathrone BattleOps: Week 2 Assignments

ASSIGNMENT 1: Control-Plane Meltdown Inspector

Goal: Diagnose API server slowness caused by etcd latency & CRDs.

PART A: Setup

1. Create cluster via kubeadm
 2. Apply Flannel
 3. Install metrics-server
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PART B: Inject Meltdown

```
for i in {1..800}; do  
  kubectl apply -f https://k8s.io/examples/application/crd.yaml  
done
```

PART C: Debug

Step 1: Check API server livez

```
kubectl get --raw '/livez?verbose'
```

Step 2: Check etcd latency

```
ETCDCTL_API=3 etcdctl endpoint status --write-out=table
```

Step 3: Observe client-side impact

```
kubectl get pods -A --request-timeout=1s
```

Step 4: Check scheduler starving

```
kubectl get events --sort-by=.metadata.creationTimestamp | tail
```

PART D: Fix

Delete CRDs:

```
for i in {1..800}; do  
  kubectl delete crd crd-$i || true  
Done
```

ASSIGNMENT 2: Scheduler Deadlock Resolver

Goal: Debug Pending pods caused by taints, affinity & resource pressure.

PART A: Inject failure

```
kubectl taint nodes <node> env=prod:NoSchedule
```

PART B: Deploy workload

```
kubectl apply -f nginx-deployment.yaml
```

```
kubectl describe pod <pod>
```

Expected: **Pod Pending**

PART C: Remove deadlock

Add toleration:

tolerations:

- key: "env"
 - operator: "Equal"
 - value: "prod"
 - effect: "NoSchedule"

PART D: Affinity conflict

Create a deployment with strict affinity:

requiredDuringSchedulingIgnoredDuringExecution:

- labelSelector:
 - matchExpressions:
 - key: zone
 - operator: In
 - values: ["us-east-5"]

Since no such node exists → pod stays Pending.

ASSIGNMENT 3: DNS vs TCP Failure Debugging

Goal: Learn difference between:

- DNS resolution
 - ICMP path
 - TCP path
 - kube-proxy behavior
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PART A: Create DNS Storm

for i in {1..3000}; do dig test.local & done

PART B: Observe kube-dns crash

kubectl get pods -n kube-system -w

PART C: Compare ping, dig, curl

ping google.com

dig google.com

curl -I https://google.com

Expected Output:

- ping → OK
- dig → OK
- curl → intermittent failures

PART D: Debug conntrack

sudo conntrack -S

Check for:

- huge "insert_failed"
- high "entries"

PART E: Fix

Restart DNS: kubectl rollout restart deployment coredns -n kube-system